SOFTWARE REQUIREMENTS SPECIFICATION

Vacation Home Insurance Management System

IEEE-aligned consolidated requirements baseline

Document ID	DOC-SRS-007
Version	1.0
Status	Consolidated SRS Baseline
Project Type	Full-Stack Insurance Capstone
Implementation Constraint	Approximately one week

Document Purpose

This document consolidates the approved requirements, use cases, business rules, user journeys, architecture, domain model, database design, and API specification into a single implementation-ready baseline.

Scope discipline: Unresolved insurance parameters and workflow decisions remain explicitly marked as TBD. They must not be silently implemented as confirmed requirements.

1. Introduction

1.1 Purpose

This Software Requirements Specification consolidates the approved requirements, use cases, business rules, user journeys, architecture, domain model, database design, and API design for the Vacation Home Insurance Management System.

It provides a single authoritative reference for:

- Project scope and boundaries
- Functional and non-functional requirements
- Actor responsibilities and permissions
- Insurance business rules
- User journeys and workflows
- Service and module ownership
- Domain entities and data relationships
- API capabilities
- Validation, security, and testing expectations
- Traceability from requirement to test

The document is implementation-ready but preserves unresolved business decisions as TBD rather than inventing unsupported insurance rules.

1.2 Scope

The system is a simplified digital platform through which a Customer can:

1. Register and authenticate.
2. View and maintain a profile.
3. Register and maintain vacation properties.
4. Select supported insurance coverages.
5. Request and view a system-calculated Quote.
6. Create and view a Policy through the Quote-to-Policy workflow.
7. Submit a Claim against an eligible Active Policy.
8. View Claim status and outcome.

A Claims Manager can authenticate, view Claims requiring review, review relevant information, and approve or reject a Claim where permitted.

An Admin can authenticate, access a protected Admin area, and view a lightweight dashboard. Claims Manager management and additional system information are optional and must not delay the core workflow.

The system is an academic capstone and not a complete commercial insurer platform.

1.3 Business Context

Vacation properties may differ from a primary residence in occupancy, exposure, security, maintenance, and disaster risk. The application demonstrates how Customer and property data progress through meaningful insurance stages:

```
Customer
-> Vacation Property
-> Coverage Selection
-> Risk Assessment
-> Premium Calculation
-> Quote
-> Policy
-> Claim
-> Claims Manager Review
-> Approved or Rejected
```

The application must demonstrate relationships and lifecycle rules rather than isolated CRUD screens.

1.4 Intended Audience

- Project evaluator
- Business analyst
- Solution architect
- Angular developer
- .NET developer
- Database developer
- Test engineer
- Security reviewer
- Future maintainer

1.5 Definitions

Term	Definition
Vacation Property	A Customer-owned property registered as the subject of insurance activity
Coverage	An approved category of protection selected for a Quote and preserved in a Policy
Risk Assessment	System-controlled evaluation that produces a Risk Score and Risk Level
Premium	Calculated price based on property value, risk, and selected Coverage
Quote	A calculated insurance offer; it is not a Policy
Policy	Insurance record created from an applicable Quote
Claim	Customer-submitted request associated with an eligible Policy
Claim Review	Claims Manager activity that moves a Claim through review to an Approved or Rejected outcome
Ownership Check	Backend verification that a Customer may access the requested record
Aggregate	A consistency boundary controlled through one aggregate root
Bounded Context	Domain boundary owned by one service

1.6 Acronyms

Acronym	Meaning
API	Application Programming Interface
EF Core	Entity Framework Core
JWT	JSON Web Token
MVP	Minimum Viable Product
RBAC	Role-Based Access Control
REST	Representational State Transfer
SRS	Software Requirements Specification
SQL	Structured Query Language
UI	User Interface
UTC	Coordinated Universal Time

1.7 References

This SRS consolidates the following project artifacts:

9. DOC-REQ-001, Requirements & Scope, Version 2.0.
10. DOC-UCUS-002, Use Cases & User Stories, Version 1.0.
11. DOC-BR-003, Business Rules, Version 1.0.
12. DOC-ARCH-004, Architecture Document.
13. DOC-DOMDB-005, Domain Model and Database Design.
14. DOC-API-006, API Specification.
15. User Journey baseline developed from the same sources.

2.1 Product Perspective

The system is a greenfield full-stack application consisting of:

```
Angular Frontend
          |
          v
```

```
Ocelot API Gateway
          |
          +-- Auth Service
          +-- Customer Service
          +-- Insurance Service
```

Each business service owns a separate SQL Server database. The Insurance Service contains exactly three internal modules:

16. Quote & Underwriting Module
17. Policy Module
18. Claims Module

2.2 Product Functions

Customer Functions

- Register
- Login and logout
- View and update own profile
- Register, view, and update own vacation properties
- View approved Coverage options
- Request and view own Quotes
- Create a Policy from an applicable Quote
- View own Policies
- Submit a Claim against an eligible Active Policy
- View own Claims, status, and final outcome

Claims Manager Functions

- Login and logout
- View Claims requiring review
- View relevant Claim, Policy, Property, and Coverage information
- Begin Claim review
- Approve or reject an Under Review Claim

Admin Functions

- Login and logout
- Access protected Admin functionality
- View a lightweight dashboard
- Optionally manage Claims Manager accounts
- Optionally view approved system information

System Functions

- Validate input
- Authenticate and authorize actors
- Enforce ownership
- Calculate Risk Score and Risk Level
- Calculate Premium
- Generate Quote

- Maintain Policy and Claim lifecycles
- Return controlled errors
- Log important events and failures
- Support automated tests

2.3 User Classes

Customer

Primary insurance user. The Customer manages only their own profile, properties, Quotes, Policies, and Claims.

Claims Manager

Authorized Claim-review user. Claims Manager permissions are distinct from Customer permissions and do not provide unrestricted Customer operations.

Admin

Protected administrative user with intentionally limited capstone scope. Admin must not be assumed to have Claim-decision authority.

2.4 Operating Environment

Client

- Modern web browser
- Angular single-page application
- Responsive desktop-oriented interface; mobile responsiveness is desirable but not a separate business capability

Server

- .NET 8 runtime
- ASP.NET Core Web APIs
- Ocelot API Gateway
- SQL Server
- Entity Framework Core

Development and Demonstration

- Local or controlled capstone environment
- HTTPS recommended; development certificates acceptable locally
- One SQL Server instance may host three independent service databases

2.5 Assumptions

19. The capstone uses a simplified deterministic risk model.
20. No external property or insurer data provider is required.
21. The five approved Coverage types are sufficient for the MVP.
22. One complete vertical slice is prioritized over optional features.
23. Claims Manager is the only confirmed Claim reviewer.

24. Capstone-only pricing parameters may be configured after approval and must not be represented as real insurer rates.
25. Basic Admin functionality remains intentionally small.
26. The application uses synchronous REST for the MVP.

2.6 Constraints

27. Implementation time is approximately one week.
28. Angular is required for the frontend.
29. .NET 8 ASP.NET Core Web APIs are required.
30. Ocelot is required as the API Gateway.
31. The solution must demonstrate microservices.
32. Each service must follow Clean Architecture principles.
33. EF Core and SQL Server are required.
34. Exactly three business roles exist.
35. Exactly three business services exist.
36. Insurance Service contains exactly three internal modules.
37. Optional features must not delay the core workflow.
38. Unnecessary enterprise infrastructure is prohibited.

2.7 Dependencies

- Angular frontend depends on Ocelot routes.
- Protected APIs depend on Auth Service JWT issuance and validation configuration.
- Customer Profile and Property operations depend on Customer Service.
- Quote generation depends on Customer Service property data.
- Policy creation depends on an applicable Quote.
- Claim submission depends on an owned Active Policy.
- Claims Manager decision depends on a Claim being Under Review.

3. Business Overview

3.1 Vacation Home Insurance

For this project, vacation home insurance is a simplified insurance product for a Customer-owned vacation property. The application evaluates approved property characteristics, allows Coverage selection, calculates Risk and Premium, and produces a Quote that may create a Policy.

3.2 Coverage

The approved Coverage catalogue contains:

- Fire
- Theft
- Natural Disaster
- Personal Liability
- Personal Belongings

At least one Coverage is required. Coverage affects Premium but does not let the Customer choose the system-calculated Risk.

3.3 Risk Assessment

Risk Assessment is automated. It may use approved property information such as value, age, type, occupancy, condition, safety and security characteristics, pool, disaster exposure, and previous claims.

The output is:

- Risk Score
- Risk Level

The Customer cannot manually set or override either output.

3.4 Premium

The confirmed conceptual formula is:

```
Base Premium = Property Value x Base Rate
Risk Adjusted Premium = Base Premium x Risk Factor
Final Premium = Risk Adjusted Premium + Coverage Charges
```

Exact rates, multipliers, charges, currency, and rounding remain TBD.

3.5 Quote

A Quote is a calculated offer containing the applicable Property reference or snapshot, selected Coverages, Risk result, Premium calculation, and status. A Quote is separate from a Policy.

Example: A Customer selects Fire and Natural Disaster Coverage for a registered property. The system evaluates risk, calculates Premium, and generates a Quote. No Policy exists until the Customer proceeds through Quote-to-Policy conversion.

3.6 Policy

A Policy is created only from an applicable Quote.

Lifecycle:

```
Pending -> Active -> Expired
                  -> Cancelled
```

Only Active Policies may receive new Claims. Renewal is outside the MVP.

3.7 Claim

A Claim is associated with one Policy and one Customer. A Customer may submit a Claim only against an authorized Active Policy.

Lifecycle:

```
Submitted -> Under Review -> Approved
                          -> Rejected
```

Approved and Rejected are final under the normal workflow.

3.8 Claims Manager Review

A Claims Manager views reviewable Claims, examines approved Claim, Policy, Property, and Coverage information, moves a Submitted Claim to Under Review, and records an Approved or Rejected decision where the finalized criteria permit it.

The current MVP does not include settlement, appeal, reopening, or multi-level investigation.

4. Business Requirements

4.1 Customer Business Requirements

ID	Business Requirement	Priority	Status
BRQ-CUS-001	The system shall allow a person to register as a Customer.	Must	Confirmed
BRQ-CUS-002	The system shall allow a Customer to authenticate and access Customer functions.	Must	Confirmed
BRQ-CUS-003	The Customer shall view and maintain their own profile.	Must	Confirmed
BRQ-CUS-004	The Customer shall register and view owned Vacation Properties.	Must	Confirmed
BRQ-CUS-005	The Customer should maintain applicable property information.	Should	Confirmed
BRQ-CUS-006	The Customer shall request and view Quotes for owned eligible properties.	Must	Confirmed
BRQ-CUS-007	The Customer shall create a Policy from an applicable Quote.	Must	Confirmed
BRQ-CUS-008	The Customer shall view their own Policies.	Must	Confirmed
BRQ-CUS-009	The Customer shall submit a Claim against an owned eligible Active Policy.	Must	Confirmed
BRQ-CUS-010	The Customer shall view their own Claims and Claim status.	Must	Confirmed

4.2 Claims Manager Business Requirements

ID	Business Requirement	Priority	Status
BRQ-AGT-001	The Claims Manager shall authenticate before accessing review functions.	Must	Confirmed
BRQ-AGT-002	The Claims Manager shall view Claims requiring review.	Must	Confirmed
BRQ-AGT-003	The Claims Manager shall review relevant Claim information.	Must	Confirmed
BRQ-AGT-004	The Claims Manager shall approve or reject an Under Review Claim where permitted.	Must	Confirmed

4.3 Admin Business Requirements

ID	Business Requirement	Priority	Status
BRQ-ADM-001	The Admin shall authenticate and access a protected Admin area.	Must	Confirmed
BRQ-ADM-002	The system shall provide a lightweight Admin dashboard or overview.	Must	Proposed scope
BRQ-ADM-003	The Admin should manage approved Claims Manager-account operations.	Should	Optional
BRQ-ADM-004	The Admin should view approved system information.	Should	Optional
BRQ-ADM-005	The Admin may view basic monitoring or reports.	Could	Optional

4.4 System Business Requirements

ID	Business Requirement	Priority	Status
BRQ-SYS-001	The system shall enforce authentication for protected operations.	Must	Derived
BRQ-SYS-002	The system shall enforce role authorization.	Must	Confirmed
BRQ-SYS-003	The system shall prevent cross-customer access.	Must	Derived
BRQ-SYS-004	The system shall calculate Risk.	Must	Confirmed
BRQ-SYS-005	The system shall calculate Premium deterministically.	Must	Confirmed
BRQ-SYS-006	The system shall enforce Policy and Claim lifecycles.	Must	Confirmed
BRQ-SYS-007	The system shall preserve data integrity and controlled errors.	Must	Derived

5. Functional Requirements

5.1 Authentication

FR-AUTH-001 Customer Registration

- **Description:** The system shall allow a person to register as a Customer.
- **Actor:** Customer
- **Priority:** Must
- **Dependencies:** Auth Service, Customer Service onboarding
- **Acceptance Criteria:**

- Valid information creates a Customer-role account.

- Invalid information is rejected.

- Password is securely hashed.

- Client cannot self-select Claims Manager or Admin.

- Sensitive values are not returned or logged.

FR-AUTH-002 User Login

- **Description:** The system shall authenticate Customer, Claims Manager, and Admin accounts.
- **Actor:** Customer, Claims Manager, Admin
- **Priority:** Must
- **Dependencies:** Existing eligible account
- **Acceptance Criteria:**

- Valid credentials produce an authenticated context.

- Invalid credentials do not grant access.

- Account role is included in authorization context.

- Ineligible accounts cannot continue.

FR-AUTH-003 Protected Access

- **Description:** Protected functions shall require valid authentication.
- **Actor:** All
- **Priority:** Must
- **Acceptance Criteria:** Missing, invalid, or expired authentication is denied.

FR-AUTH-004 Role Authorization

- **Description:** Protected functions shall be restricted according to Customer, Claims Manager, and Admin roles.
- **Priority:** Must
- **Acceptance Criteria:** Wrong-role requests return access denied without protected data.

FR-AUTH-005 Logout

- **Description:** An authenticated user shall be able to end the active client session.
- **Priority:** Must
- **Acceptance Criteria:** Subsequent protected use requires authentication again.

5.2 Customer Profile

FR-CUST-001 View Profile

- **Actor:** Customer
- **Priority:** Must
- **Dependencies:** Authentication and Customer context
- **Acceptance Criteria:** Customer views only their own profile.

FR-CUST-002 Update Profile

- **Actor:** Customer
- **Priority:** Must
- **Acceptance Criteria:**

- Customer updates only maintainable fields.

- Valid changes persist.

- Invalid changes are rejected.

- Another Customer's profile cannot be modified.

FR-CUST-003 Profile Completeness

- **Description:** Required profile information shall be available before dependent operations that need it.
- **Status:** Derived; exact fields TBD.

5.3 Vacation Property

FR-PROP-001 Register Property

- **Actor:** Customer
- **Priority:** Must
- **Acceptance Criteria:**

- Property is associated with authenticated Customer.

- Property value is greater than zero.

- Construction year is not in the future.

- Required risk information is validated.

FR-PROP-002 View Properties

- **Actor:** Customer
- **Priority:** Must
- **Acceptance Criteria:** Customer sees only their properties; an empty list is valid.

FR-PROP-003 Update Property

- **Actor:** Customer
- **Priority:** Should
- **Acceptance Criteria:**

- Customer updates only an owned property.

- Validation rules are enforced.

- Risk-relevant changes are identified.

- Existing Quotes are not silently rewritten.

5.4 Coverage

FR-COV-001 List Coverages

- **Actor:** Customer
- **Priority:** Must
- **Acceptance Criteria:** Only approved active Coverage types are shown.

FR-COV-002 Select Coverage

- **Actor:** Customer
- **Priority:** Must
- **Acceptance Criteria:**

- At least one Coverage is selected.
- Unsupported or duplicate Coverage is rejected.
- Coverage contributes to Premium.

5.5 Risk Assessment

FR-RISK-001 Calculate Risk

- **Actor:** System
- **Priority:** Must
- **Dependencies:** Eligible property and complete required risk data
- **Acceptance Criteria:**

- System produces Risk Score and Risk Level.
- Disaster exposure participates.
- Customer cannot submit or override risk outcome.
- Missing mandatory inputs block valid assessment.

FR-RISK-002 Recalculate Risk

- **Description:** Material pre-Quote input changes shall require a current risk result.
- **Priority:** Must

5.6 Premium Calculation

FR-PREM-001 Calculate Premium

- **Actor:** System
- **Priority:** Must
- **Acceptance Criteria:**

- Calculation uses Property Value, Risk Factor, and Coverage Charges.
- Identical approved inputs produce identical results.
- Calculation version is traceable.

5.7 Quote

FR-QUOTE-001 Request Quote

- **Actor:** Customer
- **Priority:** Must
- **Dependencies:** Owned eligible Property, Coverage selection, Risk and Premium configuration
- **Acceptance Criteria:**

- Ownership and eligibility are validated.
- Risk and Premium are calculated.
- Quote is persisted with calculation snapshots.
- Quote is separate from Policy.

FR-QUOTE-002 View Quote

- **Actor:** Customer
- **Priority:** Must
- **Acceptance Criteria:** Customer sees only authorized Quote information.

FR-QUOTE-003 Validate Applicability

- **Description:** A materially stale or otherwise inapplicable Quote shall not create a Policy.
- **Priority:** Must
- **Open Dependency:** Quote validity and acceptance mechanics.

5.8 Policy

FR-POL-001 Create Policy

- **Actor:** Customer
- **Priority:** Must
- **Dependencies:** Applicable owned Quote
- **Acceptance Criteria:**

- Policy references source Quote, Customer, and Property.

- Policy contains at least one Coverage.

- Policy begins in Pending.

- Repeated conversion of the same Quote is prevented in the proposed MVP design.

FR-POL-002 View Policy

- **Actor:** Customer
- **Priority:** Must
- **Acceptance Criteria:** Customer sees only their Policies.

FR-POL-003 Maintain Lifecycle

- **Actor:** System or finalized authorized actor
- **Priority:** Must
- **Acceptance Criteria:**

- Pending may become Active after finalized activation conditions.

- Active may become Expired.

- Active may become Cancelled under finalized conditions.

- Reverse transitions are rejected.

5.9 Claims

FR-CLM-001 Submit Claim

- **Actor:** Customer
- **Priority:** Must
- **Dependencies:** Owned Active Policy
- **Acceptance Criteria:**

- Policy exists and belongs to Customer.
- Policy is Active.
- Incident description is provided.
- Valid Claim starts in Submitted.

FR-CLM-002 View Claims

- **Actor:** Customer
- **Priority:** Must
- **Acceptance Criteria:** Customer sees only their Claims and current status.

FR-CLM-003 Begin Review

- **Actor:** Claims Manager
- **Priority:** Must
- **Acceptance Criteria:**

- Claims Manager role is required.
- Submitted Claim becomes Under Review.
- Finalized Claim cannot re-enter review.

FR-CLM-004 Decide Claim

- **Actor:** Claims Manager
- **Priority:** Must
- **Acceptance Criteria:**

- Claim must be Under Review.
- Decision is Approved or Rejected.
- Result is persisted and visible to Customer.
- Repeated finalization is rejected.

5.10 Administration

FR-ADM-001 Protected Admin Area

- **Actor:** Admin
- **Priority:** Must
- **Acceptance Criteria:** Non-Admin users are denied.

FR-ADM-002 Dashboard

- **Actor:** Admin
- **Priority:** Must
- **Status:** Scope proposed; metrics TBD.

FR-ADM-003 Claims Manager Management

- **Actor:** Admin
- **Priority:** Should

- **Status:** Optional; exact actions TBD.

6. User Stories

US-AUTH-001 Customer Registration

As a Customer, I want to register for an account so that I can access the insurance application.

Acceptance criteria:

- Required information is validated.
- Valid registration creates a Customer account.
- Invalid registration is rejected.
- Sensitive information is protected.

US-AUTH-002 User Login

As a Customer, Claims Manager, or Admin, I want to log in so that I can access functions permitted to my role.

Acceptance criteria:

- Valid credentials grant role-appropriate access.
- Invalid credentials do not grant access.
- Backend authorization respects the assigned role.

US-CUST-001 View Profile

As a Customer, I want to view my profile so that I can verify the information associated with my insurance activity.

US-CUST-002 Update Profile

As a Customer, I want to update maintainable profile information so that it remains current.

US-PROP-001 Register Property

As a Customer, I want to register a Vacation Property so that I can use it in the insurance workflow.

US-PROP-002 View Properties

As a Customer, I want to view my properties so that I can select the correct property for insurance activity.

US-PROP-003 Update Property

As a Customer, I want to update applicable property information so that insurance evaluation uses current information.

US-QUOTE-001 Request Quote

As a Customer, I want to request a Quote for an owned eligible property so that I can receive insurance pricing information.

Acceptance criteria:

- At least one approved Coverage is selected.
- Risk is calculated by the system.
- Premium is calculated using approved inputs.
- Unauthorized or incomplete Property data is rejected.

US-QUOTE-002 View Quote

As a Customer, I want to view my Quote so that I can understand the calculated offer.

US-POL-001 Create Policy

As a Customer, I want to create a Policy from an applicable Quote so that the supported insurance workflow produces a Policy.

US-POL-002 View Policy

As a Customer, I want to view my Policy so that I can understand its Coverage and status.

US-CLM-001 Submit Claim

As a Customer, I want to submit a Claim against an eligible Active Policy so that it can enter review.

US-CLM-002 Track Claim

As a Customer, I want to view my Claim status so that I can understand its progress and outcome.

US-CLM-003 Review Claim

As a Claims Manager, I want to review Claims requiring attention so that I can perform the Claim-review function assigned to my role.

US-CLM-004 Record Decision

As a Claims Manager, I want to approve or reject an Under Review Claim where permitted so that the review reaches a final outcome.

US-ADM-001 Access Admin Area

As an Admin, I want to access protected administration so that I can perform approved system-level functions.

US-ADM-002 View Dashboard

As an Admin, I want to view a lightweight system overview so that I have basic operational visibility.

7. Use Cases

UC-AUTH-001 Register Customer

- **Actor:** Customer
- **Goal:** Establish a Customer account.
- **Preconditions:** Registration information can be supplied.
- **Postconditions:** Account and Customer context are created, or registration is rejected.

Main Flow

39. Customer opens registration.
40. Customer enters required information.
41. System validates the request.
42. System creates a Customer-role User.
43. System establishes the Customer context.
44. System confirms registration.

Alternative Flow

- Customer corrects invalid input and resubmits.

Exception Flow

- Duplicate login identifier.
- Customer Service unavailable during onboarding.
- Persistence failure.

UC-AUTH-002 Login

- **Actor:** Customer, Claims Manager, Admin
- **Goal:** Authenticate and obtain role-appropriate access.
- **Preconditions:** Eligible account exists.
- **Postconditions:** Authenticated context is created or access is denied.

Main Flow

45. User submits credentials.
46. System verifies password.
47. System checks account eligibility.
48. System identifies role.
49. System issues JWT.
50. UI routes user to permitted functionality.

UC-CUST-001 Manage Profile

- **Actor:** Customer
- **Goal:** View or update own profile.
- **Preconditions:** Customer is authenticated.
- **Postconditions:** Profile is displayed or valid changes are stored.

Main Flow

51. Customer requests profile.
52. System derives Customer identity from JWT.
53. System returns own profile.
54. Customer edits permitted fields.
55. System validates and stores changes.

Exception Flow

- Profile unavailable.
- Invalid field.
- Stale concurrency value.

UC-PROP-001 Register Property

- **Actor:** Customer
- **Goal:** Register a property for insurance activity.
- **Preconditions:** Customer is authenticated.
- **Postconditions:** Valid property is associated with Customer.

Main Flow

56. Customer enters property information.
57. System validates ownership context.
58. System validates Property Value and Construction Year.
59. System validates required risk data.
60. System stores Property.

Alternative Flow

- Customer corrects invalid fields.

UC-QUOTE-001 Generate Quote

- **Actor:** Customer
- **Goal:** Obtain a calculated Quote.
- **Preconditions:** Customer owns a Property suitable for evaluation.
- **Postconditions:** Quote is generated or request is rejected.

Main Flow

61. Customer selects owned Property.
62. Customer selects at least one Coverage.
63. Insurance Service obtains authoritative Property data.
64. System validates eligibility.
65. System calculates Risk Score and Risk Level.
66. System calculates Premium.
67. System stores Quote snapshots.
68. System returns Quote.

Alternative Flow

- Missing data requires Property correction.
- Material change requires new calculation.

Exception Flow

- Unauthorized Property.
- Unsupported Coverage.
- Customer Service unavailable.

UC-POL-001 Create Policy

- **Actor:** Customer
- **Goal:** Create a Policy from an applicable Quote.
- **Preconditions:** Applicable owned Quote exists.
- **Postconditions:** Pending Policy is created or conversion is rejected.

Main Flow

69. Customer selects Quote.
70. System verifies ownership.
71. System validates applicability.
72. System copies Coverage, Risk, and Premium snapshots.
73. System creates Pending Policy.
74. System returns Policy.

Exception Flow

- Quote already used.
- Quote stale or inapplicable.
- Required Policy rule unresolved or misconfigured.

UC-CLM-001 Submit Claim

- **Actor:** Customer
- **Goal:** Submit Claim against an eligible Policy.
- **Preconditions:** Owned Active Policy exists.
- **Postconditions:** Claim is Submitted or rejected.

Main Flow

75. Customer selects Policy.
76. System validates ownership and Active status.
77. Customer supplies required Claim information.
78. System validates the request.
79. System creates Claim in Submitted.

UC-CLM-003 Review Claim

- **Actor:** Claims Manager
- **Goal:** Review a Claim requiring attention.
- **Preconditions:** Claims Manager is authenticated; Submitted Claim exists.
- **Postconditions:** Claim is Under Review.

Main Flow

80. Claims Manager opens review queue.
81. System returns reviewable Claims.
82. Claims Manager selects Claim.
83. System displays approved review information.
84. Claims Manager begins review.
85. System changes Submitted to Under Review.

UC-CLM-004 Record Claim Decision

- **Actor:** Claims Manager
- **Goal:** Approve or reject a Claim.
- **Preconditions:** Claim is Under Review.
- **Postconditions:** Claim is Approved or Rejected.

Main Flow

86. Claims Manager completes review.
87. Claims Manager selects Approved or Rejected.
88. System validates current state and authority.
89. System records decision.
90. Customer-visible status is updated.

UC-ADM-001 View Admin Dashboard

- **Actor:** Admin
- **Goal:** Access protected administration and basic overview.
- **Preconditions:** Admin is authenticated.
- **Postconditions:** Approved overview is displayed.

Exception Flow

- Non-Admin access is denied.
- Optional data is unavailable.

8. Business Rules

8.1 Authentication and Authorization Rules

91. Registration creates a Customer account.
92. Authentication is required for protected operations.
93. Role determines business access.
94. Account must be eligible to authenticate.
95. Logout ends the active client authentication context.
96. Exactly three roles exist: Customer, Claims Manager, Admin.
97. Admin Claim-decision authority is TBD and denied by default.

8.2 Customer Rules

98. Customer profile belongs to that Customer.

99. Customer cannot access another Customer's protected information.
100. Required profile information must exist before dependent workflows use it.
101. Customer may update only designated fields.

8.3 Property Rules

102. Property requires a Customer association.
103. Only authenticated Customer may register a Property.
104. Required evaluation information must be available.
105. Property Value must be greater than zero.
106. Construction Year cannot be in the future.
107. Multiple properties are permitted.
108. Customer may act only on owned properties.
109. Eligibility is required before Quote generation.
110. Risk-relevant changes must be considered by later evaluation.

8.4 Coverage Rules

111. Only approved Coverage types may be selected.
112. At least one Coverage is required.
113. Coverage affects Premium.
114. Coverage belongs to a Quote or Policy context.
115. Coverage does not allow Customer-selected Risk.
116. Coverage limits remain TBD.

8.5 Risk Rules

117. Risk is system-calculated.
118. Risk uses approved property information.
119. Disaster exposure participates.
120. Disaster exposure is explicit Yes or No.
121. Risk produces Score and Level.
122. Missing required data blocks valid assessment.
123. Relevant changes require recalculation.
124. Customer cannot override Risk.

8.6 Quote and Premium Rules

125. Quote requires owned eligible Property.
126. Quote requires valid Coverage.
127. Risk precedes final pricing.
128. Identical approved inputs produce identical Premium.
129. Quote is separate from Policy.
130. Policy requires applicable Quote.
131. Material input changes may invalidate Quote.
132. Quote validity remains TBD.
133. Premium depends on Property Value, Risk, and Coverage.
134. Pricing parameters remain TBD until approved.

8.7 Policy Rules

135. Policy requires applicable Quote.
136. Policy requires eligible Property.
137. Policy requires at least one Coverage.
138. Policy belongs to one Customer and one Property.
139. Initial status is Pending.
140. Pending may become Active.
141. Active may become Expired.
142. Active may become Cancelled.
143. Expired or Cancelled Policies cannot accept new Claims.
144. End Date cannot precede Start Date.
145. Renewal is outside MVP.
146. Unsupported post-issuance changes are prohibited.

8.8 Claim Rules

147. Claim requires a Policy.
148. Claim belongs to the relevant Policy and Customer.
149. New Claim requires Active Policy.
150. Incident description is required.
151. Initial status is Submitted.
152. Claims Manager begins review: Submitted to Under Review.
153. Under Review may become Approved or Rejected.
154. Approved and Rejected are final.
155. Direct Submitted-to-final transition is prohibited.
156. Claim amount, date, duplicate, and resubmission rules remain unresolved.

8.9 Data Integrity Rules

157. Every Policy references one Property.
158. Every Policy has at least one Coverage.
159. Every Claim references one Policy.
160. Customer activity is associated with the correct Customer.
161. Quote and Policy snapshots remain consistent.
162. Final Claim state cannot behave as Under Review.
163. Record existence alone does not permit an action; lifecycle state must allow it.

9. User Journey

9.1 Customer Journey

Stage	Customer Action	System Action	Key Rules	Outcome
Register	Submit account information	Validate and create Customer account	Registration and role rules	Account created or rejected

Login	Submit credentials	Authenticate and issue JWT	Account eligibility	Access granted or denied
Profile	View/update own information	Enforce ownership and validation	Customer rules	Profile available/current
Property	Register or update Property	Validate value, year, risk inputs	Property rules	Property available for Quote
Coverage	Select approved Coverage	Require at least one	Coverage rules	Valid selection
Quote	Submit Property and Coverage selection	Calculate Risk and Premium	Risk, Premium, Quote rules	Quote generated
Policy	Proceed from Quote	Validate and create Pending Policy	Policy rules	Policy created
Claim	Select Active Policy and describe incident	Validate and create Submitted Claim	Claim rules	Claim enters workflow
Track	Open Claim	Return current status	Ownership	Status visible
Outcome	View final Claim	Return Approved or Rejected	Final-state rule	Outcome visible

9.2 Claims Manager Journey

```
Login
-> View review queue
-> Select Submitted Claim
-> View relevant details
-> Begin review
-> Under Review
-> Approve or Reject
```

9.3 Admin Journey

```
Login
-> Protected Admin area
-> Dashboard
-> Optional Claims Manager management
-> Optional system information
```

10. System Workflows

10.1 Customer Workflow

```
[Register]
    |
    v
[Login]
    |
    v
[Profile Complete?] -- No --> [Update Profile]
    |
   Yes
    v
[Register Property]
    |
    v
[Request Quote]
    |
    v
[Create Policy]
    |
    v
[Policy Active?] -- No --> [Claim Blocked]
    |
   Yes
    v
[Submit Claim]
    |
    v
[Track Outcome]
```

10.2 Quote Workflow

```
Owned Property
    |
    v
Eligibility and Required Data
    |
    v
At Least One Coverage
    |
    v
Risk Assessment
    |
    v
Premium Calculation
    |
    v
Quote Generated
```

10.3 Policy Workflow

```
Applicable Quote
    |
    v
Pending Policy
    |
    v
Activation Conditions Satisfied?
   / \
 No  Yes
 |  |
Wait Active
    | \
    |  \
```

```
   v   v
  Expired Cancelled
```

10.4 Claim Workflow

```
Owned Active Policy
    |
    v
Valid Claim Information
    |
    v
Submitted
    |
    v
Under Review
   / \
  v   v
Approved Rejected
```

10.5 Claims Manager Review Workflow

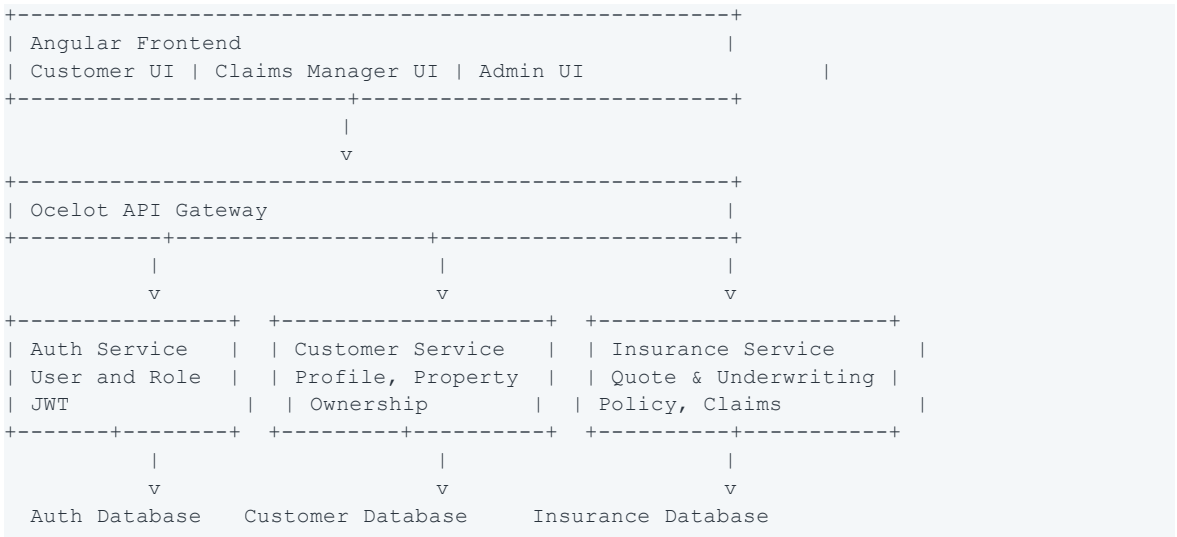
```
Claims Manager Login
-> Review Queue
-> Select Claim
-> Validate Submitted State
-> Begin Review
-> Review Information
-> Record Approved or Rejected
```

10.6 Admin Workflow

```
Admin Login
-> Role Validation
-> Protected Dashboard
-> Optional Approved Administration
```

11. System Architecture

11.1 High-Level Architecture



11.2 Service Responsibilities

API Gateway

- External entry point
- Route requests
- Forward JWT and correlation ID
- Safe request metadata logging
- Optional rate limiting

Auth Service

- Registration
- Login
- Password security
- JWT
- Roles and account eligibility

Customer Service

- Customer
- Customer Profile
- Vacation Property
- Ownership validation
- Property evaluation data

Insurance Service

Quote & Underwriting Module

- Coverage
- Property eligibility
- Risk
- Premium
- Quote

Policy Module

- Quote-to-Policy conversion
- Policy lifecycle
- Policy retrieval

Claims Module

- Claim submission and tracking
- Claims Manager review
- Approval and rejection

11.3 Communication

- Angular to Gateway: HTTPS REST/JSON.
- Gateway to Services: synchronous REST.

- Insurance to Customer Service: synchronous REST for authoritative Property data.
- Insurance modules: in-process method calls and interfaces.
- No message broker or distributed transaction.

12. Domain Model

12.1 Entities

Auth Context

- User
- Role
- UserRole association

Customer Context

- Customer
- CustomerProfile
- VacationProperty

Insurance Context

- Coverage
- RiskAssessment
- Quote
- QuoteCoverage
- Policy
- PolicyCoverage
- Claim
- ClaimReview

12.2 Aggregate Roots

```
User
Customer
VacationProperty
Quote
Policy
Claim
```

12.3 Relationships

```
User 1 ---- 0..1 Customer
Customer 1 ---- 1 CustomerProfile
Customer 1 ---- 0..* VacationProperty
VacationProperty 1 ---- 0..* Quote      logical cross-service reference
Quote 1 ---- 1 RiskAssessment
Quote 1 ---- 1..* QuoteCoverage
Quote 1 ---- 0..1 Policy
Policy 1 ---- 1..* PolicyCoverage
Policy 1 ---- 0..* Claim
Claim 1 ---- 0..1 ClaimReview
```

12.4 Aggregate Invariants

- Quote controls Coverage selection, Risk result, Premium, and snapshots.
- Policy controls Coverage at issuance and lifecycle.
- Claim controls review and final decision.
- Child entities are not modified independently through public APIs.

13. Database Design

13.1 Strategy

Use one SQL Server instance with three independent databases:

```
VacationHomeAuthDb
VacationHomeCustomerDb
VacationHomeInsuranceDb
```

There are no cross-database foreign keys.

13.2 Auth Database

Tables:

- Users
- Roles
- UserRoles
- AuditLogs, optional

Important constraints:

- Unique normalized email
- Recognized roles only
- One role per User in MVP

13.3 Customer Database

Tables:

- Customers
- CustomerProfiles
- VacationProperties
- AuditLogs, optional

Important constraints:

- Unique AuthUserId
- One Profile per Customer
- Positive Property Value
- Non-negative previous Claim count
- Customer ownership foreign key

13.4 Insurance Database

Tables:

- Coverages
- Quotes
- QuoteCoverages

- RiskAssessments
- Policies
- PolicyCoverages
- Claims
- ClaimReviews
- AuditLogs, optional

Important constraints:

- Unique Coverage Code
- Unique Quote, Policy, and Claim numbers
- One RiskAssessment per Quote
- One Policy per Quote in proposed MVP
- One ClaimReview per Claim
- Constrained lifecycle status values

13.5 Indexes

Recommended indexes:

```
Users(NormalizedEmail) unique
Customers(AuthUserId) unique
VacationProperties(CustomerId)
Quotes(CustomerId, CreatedAt)
Quotes(VacationPropertyId, CreatedAt)
Policies(CustomerId, Status)
Policies(VacationPropertyId, Status)
Claims(CustomerId, SubmittedAt)
Claims(PolicyId, SubmittedAt)
Claims(Status, SubmittedAt)
ClaimReviews(Claims ManagerUserId, ReviewStartedAt)
```

13.6 ER Diagram

```
Users -- UserRoles -- Roles
  |
  | logical AuthUserId
  v
Customers -- CustomerProfiles
  |
  +-- VacationProperties
           |
           | logical PropertyId
           v
Quotes -- QuoteCoverages -- Coverages
  |
  +-- RiskAssessments
  |
  +-- Policies -- PolicyCoverages -- Coverages
           |
           +-- Claims -- ClaimReviews
```

14. API Specification Summary

All external APIs use `/api/v1` and pass through Ocelot.

14.1 Auth APIs

Method	Path	Access
POST	`/api/v1/auth/register`	Public
POST	`/api/v1/auth/login`	Public
POST	`/api/v1/auth/logout`	Authenticated
GET	`/api/v1/auth/me`	Authenticated

14.2 Customer APIs

Method	Path	Access
GET	`/api/v1/customers/profile`	Customer
PUT	`/api/v1/customers/profile`	Customer

14.3 Property APIs

Method	Path	Access
POST	`/api/v1/properties`	Customer
GET	`/api/v1/properties`	Customer
GET	`/api/v1/properties/{id}`	Customer, own only
PUT	`/api/v1/properties/{id}`	Customer, own only

Property deletion is excluded because no approved use case exists.

14.4 Quote APIs

Method	Path	Access
GET	`/api/v1/coverages`	Customer

POST	`/api/v1/quotes`	Customer
GET	`/api/v1/quotes`	Customer, own only
GET	`/api/v1/quotes/{id}`	Customer, own only

Risk Assessment is not exposed as a Customer-controlled endpoint. It occurs within Quote generation.

14.5 Policy APIs

Method	Path	Access
POST	`/api/v1/policies`	Customer
GET	`/api/v1/policies`	Customer, own only
GET	`/api/v1/policies/{id}`	Customer, own only

Activation and cancellation action endpoints remain reserved until business rules and authorized actors are finalized.

14.6 Claim and Claims Manager APIs

Method	Path	Access
POST	`/api/v1/claims`	Customer
GET	`/api/v1/claims`	Customer, own only
GET	`/api/v1/claims/{id}`	Customer, own only
GET	`/api/v1/Claims Manager/claims`	Claims Manager
GET	`/api/v1/Claims Manager/claims/{id}`	Claims Manager
PUT	`/api/v1/Claims Manager/claims/{id}/review`	Claims Manager

PUT	`/api/v1/Claims Manager/claims/{id}/decision`	Claims Manager

14.7 Admin APIs

Method	Path	Access	Status
GET	`/api/v1/admin/dashboard`	Admin	Minimum scope, content TBD
GET	`/api/v1/admin/system-overview`	Admin	Optional
GET	`/api/v1/admin/Claims Managers`	Admin	Optional
POST	`/api/v1/admin/Claims Managers`	Admin	Optional
PUT	`/api/v1/admin/Claims Managers/{id}`	Admin	Optional

15. Security Requirements

15.1 Authentication

164. Auth Service shall issue signed JWT access tokens.
165. Services shall validate signature, issuer, audience, and expiration.
166. Passwords shall use an approved adaptive hash.
167. Plain-text passwords shall never be stored or logged.

15.2 Authorization

168. Customer, Claims Manager, and Admin roles shall be enforced by backend policy.
169. Claims Manager Claim decisions shall require Claims Manager role.
170. Admin Claim decisions shall be denied until explicitly approved.
171. UI guards shall not be treated as security boundaries.

15.3 Ownership Validation

172. Customer identity shall come from verified JWT claims.
173. Customer-provided CustomerId shall not authorize access.
174. Customer record queries shall include authenticated Customer ownership.
175. Non-owned records may return a non-disclosing Not Found response.

15.4 Data Protection

- Use HTTPS outside local development.
- Keep secrets outside source control.
- Avoid sensitive response fields.
- Avoid unnecessary PII in logs.
- Use parameterized EF Core access.
- Protect connection strings and signing keys.

15.5 Input Validation

- Validate at UI, API/Application, and Domain layers.
- Backend validation is authoritative.
- Use allow-listed enum values.
- Enforce length, range, status, and ownership constraints.

15.6 Audit Logging

Record critical actions such as:

- Registration
- Property update
- Quote generation
- Policy creation and transition
- Claim submission
- Review start
- Claim decision

Persistent AuditLog tables are optional; structured application logging is mandatory.

16. Non-Functional Requirements

16.1 Performance

ID	Requirement
NFR-PERF-001	Core operations shall remain responsive under capstone demonstration load.
NFR-PERF-002	API and database access shall use asynchronous I/O.
NFR-PERF-003	Lists shall support pagination where record growth is plausible.
NFR-PERF-004	Optimization shall target measured bottlenecks rather than speculative infrastructure.

16.2 Availability

ID	Requirement
NFR-AVL-001	The system shall remain usable during normal demonstration and testing.
NFR-AVL-002	Failure of optional Admin functions should not block core Customer flows where practical.
NFR-AVL-003	Services should expose health endpoints.

No production availability percentage is required.

16.3 Reliability

ID	Requirement
NFR-REL-001	Unhandled failures shall produce controlled responses.
NFR-REL-002	Required service-call failures shall use timeouts and controlled handling.
NFR-REL-003	Duplicate or stale state-changing operations shall be detected where practical.
NFR-REL-004	State transitions shall use optimistic concurrency.

16.4 Maintainability

ID	Requirement
NFR-MNT-001	Each service shall follow Clean Architecture.
NFR-MNT-002	Domain logic shall remain independent of controllers and EF Core.
NFR-MNT-003	Naming, validation, logging, and errors shall follow consistent conventions.
NFR-MNT-004	The solution shall avoid infrastructure unnecessary for MVP.

16.5 Scalability

ID	Requirement
NFR-SCL-001	APIs should be stateless except for persistent business data.
NFR-SCL-002	Service boundaries shall support future independent scaling.
NFR-SCL-003	Separate databases shall preserve ownership.
NFR-SCL-004	Production-scale orchestration is not required.

16.6 Security

ID	Requirement
NFR-SEC-001	Protected operations require authentication.
NFR-SEC-002	Backend shall enforce role authorization.
NFR-SEC-003	Customer cross-access shall be prevented.
NFR-SEC-004	Sensitive values shall not be unnecessarily exposed.
NFR-SEC-005	Critical inputs shall be validated.
NFR-SEC-006	Frontend controls shall not replace backend enforcement.

16.7 Observability

ID	Requirement
NFR-OBS-001	Important events and failures shall be logged.
NFR-OBS-002	Logs shall include sufficient context and correlation IDs.
NFR-OBS-003	Credentials, tokens, and unnecessary PII shall not be logged.

16.8 Testability

ID	Requirement
NFR-TST-001	Critical domain rules shall have unit tests.
NFR-TST-002	Important use cases shall have integration tests.
NFR-TST-003	The core end-to-end workflow shall be verified.
NFR-TST-004	Tests shall be traceable to requirements.

16.9 Usability

ID	Requirement
NFR-USA-001	Core Customer workflows shall be understandable without technical knowledge.
NFR-USA-002	Errors shall explain corrective action where practical.
NFR-USA-003	Navigation shall clearly separate Customer, Claims Manager, and Admin areas.
NFR-USA-004	Claims Manager review shall present approved information needed for a decision.

17. Validation and Error Handling

17.1 Validation Layers

UI Validation

- Required-field feedback
- Format checks
- Range guidance
- Disable submit when obviously invalid

Application Validation

- Request shape
- Required fields
- Enum allow lists
- Authorization context

- External reference validation

Domain Validation

- Ownership
- Eligibility
- At least one Coverage
- System-owned Risk
- Quote applicability
- Policy and Claim transitions

Database Validation

- Primary and foreign keys within a service database
- Unique constraints
- Check constraints
- Concurrency tokens

17.2 Standard Error Response

```
{
  "success": false,
  "message": "The request could not be completed.",
  "errors": [
    {
    "code": "POLICY_NOT_ACTIVE",
    "field": "policyId",
    "message": "Only an active policy can receive a new claim."
    }
  ],
  "traceId": "correlation-value"
}
```

17.3 Status Standards

- 200 successful retrieval or action
- 201 resource created
- 400 request validation failure
- 401 unauthenticated
- 403 unauthorized role
- 404 unavailable or non-owned resource
- 409 lifecycle, duplicate, or concurrency conflict
- 503 required dependency unavailable
- 500 controlled unexpected failure

17.4 Error Principles

176. Do not expose stack traces.
177. Do not expose whether another Customer owns a record.
178. Use stable error codes.
179. Log trace ID and safe context.
180. Do not report success until persistence completes.

18. Testing Requirements

18.1 Unit Testing

Required focus:

- Property validation
- Coverage validation
- Risk calculation
- Premium calculation
- Quote consistency
- Policy transition rules
- Claim transition rules
- Customer ownership policies

18.2 Integration Testing

Required focus:

- Registration and Customer onboarding
- Login and JWT issuance
- Profile and Property persistence
- Insurance-to-Customer service call
- Quote aggregate persistence
- Quote-to-Policy conversion
- Claim submission
- Claims Manager review and final decision

18.3 API Testing

Verify:

- Request validation
- Status codes
- Response envelopes
- Authentication
- Role restrictions
- Ownership
- Pagination
- Idempotency and concurrency where implemented

18.4 Workflow Testing

Minimum happy path:

```
Register
-> Login
-> Complete Profile
-> Register Property
-> Select Coverage
-> Generate Quote
-> Create Policy
-> Activate under approved test condition
```

```
-> Submit Claim
-> Claims Manager begins review
-> Claims Manager approves or rejects
-> Customer views outcome
```

18.5 Security Testing

- Customer accesses another Customer's Property, Quote, Policy, or Claim.
- Claims Manager accesses Admin route.
- Admin attempts Claim decision.
- Expired token is used.
- JWT role is tampered with.
- Invalid input attempts injection or over-posting.
- Sensitive fields are absent from logs and responses.

18.6 Example Test Catalogue

Test ID	Requirement	Scenario	Expected Result
TC-AUTH-001	FR-AUTH-001	Valid registration	Customer account created
TC-AUTH-002	FR-AUTH-002	Invalid login	401, no access
TC-PROP-001	FR-PROP-001	Property Value zero	Validation failure
TC-PROP-002	FR-PROP-001	Future Construction Year	Validation failure
TC-QUOTE-001	FR-QUOTE-001	No Coverage selected	Quote rejected
TC-QUOTE-002	FR-RISK-001	Customer submits Risk Score	Value ignored/rejected; system calculates
TC-POL-001	FR-POL-001	Applicable Quote	Pending Policy created
TC-POL-002	FR-POL-001	Same Quote reused	Conflict
TC-CLM-001	FR-CLM-001	Active owned Policy	Submitted Claim created
TC-CLM-002	FR-CLM-001	Cancelled Policy	Claim rejected
TC-REV-001	FR-CLM-003	Claims Manager begins review	Under Review
TC-REV-002	FR-CLM-004	Under Review approved	Approved final state

TC-SEC-001	NFR-SEC-003	Cross-customer Property	Access denied/non-disclosing not found
TC-CON-001	NFR-REL-004	Two Claims Manager decisions	One succeeds, stale request conflicts

19. Risk Analysis

19.1 Business Risks

Risk	Impact	Mitigation
Risk weights unapproved	Quote outcome unstable	Isolate versioned configuration and obtain approval
Premium parameters unapproved	Premium cannot be finalized	Use configurable capstone-only values after approval
Quote validity unresolved	Policy applicability ambiguous	Keep validity nullable and rule replaceable
Activation condition unresolved	Pending Policy cannot consistently become Active	Finalize before exposing transition
Claim criteria incomplete	Claims Manager decision may be subjective	Finalize minimum decision criteria
Coverage limits unresolved	Claim amount validation incomplete	Do not invent limits
Admin boundaries unclear	Scope creep or over-authorization	Implement only protected dashboard first

19.2 Technical Risks

Risk	Impact	Mitigation
Customer Service unavailable during Quote	Quote fails	Timeout, limited retry for GET, controlled 503
Partial registration onboarding	Auth User without Customer context	Idempotent Customer creation and recovery logging

Cross-service database query	Tight coupling	Separate connections and API-only integration
Concurrent Claim decisions	Conflicting final states	RowVersion and conditional transition
Sensitive logging	Security exposure	Structured allow-list logging
Excessive architecture boilerplate	One-week delivery risk	Keep layers thin and features focused

19.3 Project Risks

Risk	Impact	Mitigation
Optional features consume schedule	Incomplete core workflow	Must-Have completion gate
Too many UI screens	Delivery delay	Reuse layouts and components
Testing deferred	Defects in final demo	Test domain rules while implementing
Unapproved assumptions embedded	Rework	Maintain TBD register and configuration boundaries

20. Traceability Matrix

20.1 Core Traceability

Requirement	User Story	Use Case	Business Rule	Entity	API	Test
FR-AUTH-001	US-AUTH-001	UC-AUTH-001	BR-AUTH-001	User, Role	POST `/auth/register`	TC-AUTH-001
FR-AUTH-002	US-AUTH-002	UC-AUTH-002	BR-AUTH-002, 004	User	POST `/auth/login`	TC-AUTH-002
FR-AUTH-004	US-AUTH-002	UC-AUTH-002	BR-AUTH-003	Role, UserRole	Protected APIs	TC-SEC-ROLE

FR-CUST-001	US-CUST-001	UC-CUST-001	BR-CUST-001 to 003	CustomerProfile	GET `/customers/profile`	TC-CUST-001
FR-CUST-002	US-CUST-002	UC-CUST-001	BR-CUST-003, 004	CustomerProfile	PUT `/customers/profile`	TC-CUST-002
FR-PROP-001	US-PROP-001	UC-PROP-001	BR-PROP-001 to 006	VacationProperty	POST `/properties`	TC-PROP-001, 002
FR-PROP-002	US-PROP-002	UC-PROP-001	BR-PROP-006, 007	VacationProperty	GET `/properties`	TC-PROP-003
FR-PROP-003	US-PROP-003	UC-PROP-001	BR-PROP-009	VacationProperty	PUT `/properties/{id}`	TC-PROP-004
FR-COV-001	US-QUOTE-001	UC-QUOTE-001	BR-COV-001 to 005	Coverage	GET `/coverages`	TC-COV-001
FR-RISK-001	US-QUOTE-001	UC-QUOTE-001	BR-RISK-001 to 008	RiskAssessment	POST `/quotes`	TC-QUOTE-002
FR-PREM-001	US-QUOTE-001	UC-QUOTE-001	BR-PREM-001 to 007	Quote	POST `/quotes`	TC-PREM-001
FR-QUOTE-001	US-QUOTE-001	UC-QUOTE-001	BR-QUOTE-001 to 007	Quote	POST `/quotes`	TC-QUOTE-001
FR-QUOTE-002	US-QUOTE-002	UC-QUOTE-001	BR-QUOTE-005	Quote	GET `/quotes/{id}`	TC-QUOTE-003
FR-POL-001	US-POL-001	UC-POL-001	BR-POL-001 to 007	Policy, PolicyCoverage	POST `/policies`	TC-POL-001, 002
FR-POL-002	US-POL-002	UC-POL-001	BR-POL-004, 005	Policy	GET `/policies/{id}`	TC-POL-003
FR-POL-003	US-POL-001	UC-POL-001	BR-POL-005 to 013	PolicyStatus	Lifecycle actions	TC-POL-STATE
FR-CLM-001	US-CLM-001	UC-CLM-001	BR-CLM-001 to 006	Claim	POST `/claims`	TC-CLM-001, 002
FR-CLM-002	US-CLM-002	UC-CLM-001	BR-CUST-002	Claim	GET `/claims/{id}`	TC-SEC-CLAIM

FR-CLM-003	US-CLM-003	UC-CLM-003	BR-CLM-007 to 009	ClaimReview	PUT `/Claims Manager/claims/{id}/review`	TC-REV-001
FR-CLM-004	US-CLM-004	UC-CLM-004	BR-CLM-010 to 012	Claim, ClaimReview	PUT `/Claims Manager/claims/{id}/decision`	TC-REV-002, TC-CON-001
FR-ADM-001	US-ADM-001	UC-ADM-001	BR-ROLE-005, 006	User, Role	`/admin/*`	TC-ADM-001
FR-ADM-002	US-ADM-002	UC-ADM-001	Limited Admin scope	Existing read data	GET `/admin/dashboard`	TC-ADM-002

20.2 Cross-Cutting Traceability

Requirement	Implementation	Verification
FR-SYS-001	UI, application, and domain validation	Validation tests
FR-SYS-002	Standard error envelope and Problem Details mapping	API contract tests
FR-SYS-003	Structured logging	Log inspection
FR-SYS-004	Central exception middleware	Exception integration tests
FR-SYS-005	Automated test suites	CI or local test run
FR-SYS-006	Typed REST clients	Integration tests
FR-SYS-007	Timeout and controlled retry	Dependency-failure tests
NFR-SEC-*	JWT, RBAC, ownership	Security tests
NFR-REL-*	Transactions and concurrency	Conflict and rollback tests
NFR-OBS-*	Correlation IDs and safe logs	Observability review

NFR-DATA-*	Constraints and aggregate invariants	Database and domain tests

21. Open Items

21.1 Customer and Property

- Exact registration fields
- Exact Customer Profile fields and validation limits
- Exact Property fields
- Property Type values
- Occupancy values
- Property Condition values
- Additional eligibility exclusions
- Minimum or maximum Property Value

21.2 Risk and Premium

- Risk factor points and weights
- Score range approval
- Risk Level thresholds
- Risk multipliers
- Base Rate
- Coverage Charges
- Currency
- Rounding
- Minimum or maximum Premium

21.3 Quote

- Validity period
- Acceptance mechanics
- Final Quote status model
- Exact behavior after material input changes

21.4 Policy

- Activation actor and condition
- Standard duration
- Cancellation actor and rules
- Multiple Active Policies per Property
- Post-issuance Coverage changes

21.5 Claim

- Required Claim fields
- Claim Amount rule
- Coverage limits

- Incident Date requirement and eligibility
- Approval and rejection criteria
- Decision-reason requirement
- Duplicate Claim behavior
- Rejected-Claim resubmission
- Any exceptional reopening rule

21.6 Admin

- Dashboard metrics
- Claims Manager-management actions
- System-overview content
- Claim-decision authority

22. Success Criteria

22.1 Business Success Criteria

The MVP is successful when:

181. Customer registers and logs in.
182. Customer views or updates own profile.
183. Customer registers and views a Vacation Property.
184. Customer selects Coverage and requests a Quote.
185. System calculates Risk and Premium.
186. Customer views the Quote.
187. Customer creates and views a Pending Policy.
188. Policy can be made Active under an approved demonstration condition.
189. Customer submits a Claim against the Active Policy.
190. Customer views Claim status.
191. Claims Manager reviews the Claim.
192. Claims Manager approves or rejects the Claim.
193. Customer views the final outcome.
194. Admin accesses protected dashboard functionality.
195. Cross-customer access is prevented.

22.2 Technical Success Criteria

- Angular application works through Ocelot.
- Auth, Customer, and Insurance Services run independently.
- Each service uses its own SQL Server database.
- EF Core migrations and persistence work.
- JWT authentication and RBAC work.
- Global exception handling and standard errors work.
- Important events and failures are logged safely.
- Critical domain logic has automated tests.
- Required service communication handles failure appropriately.

22.3 MVP Completion Gate

Must Be Complete

- Core Customer vertical slice
- Claims Manager review and decision
- Minimum Admin access
- Authentication and authorization
- Ownership
- Validation and error handling
- Database persistence
- Essential tests

Must Not Block Completion

- Claims Manager management
- System overview
- Reports
- Notifications
- Attachments
- UI polish beyond demonstrability

23. Final SRS Summary

23.1 Business Scope

The system supports the complete simplified Vacation Home Insurance workflow from Customer registration through Claims Manager Claim decision. It uses exactly three active roles and preserves the approved Coverage, Risk, Premium, Policy, and Claim rules.

23.2 System Scope

The solution consists of:

- Angular frontend
- Ocelot API Gateway
- Auth Service
- Customer Service
- Insurance Service
- Three service-owned SQL Server databases

Insurance Service contains exactly Quote & Underwriting, Policy, and Claims modules.

23.3 Domain and Data

The core domain entities are:

```
User, Role
Customer, CustomerProfile, VacationProperty
Coverage, RiskAssessment, Quote
Policy
Claim, ClaimReview
```

Aggregate and database boundaries enforce ownership and lifecycle consistency.

23.4 APIs

The API is RESTful, JSON-based, versioned under `/api/v1`, secured using JWT, and designed around business workflows rather than table CRUD. Customer identity comes from authentication claims, and Claims Manager decisions use explicit lifecycle actions.

23.5 Security

Security combines:

- Password hashing
- JWT validation
- RBAC
- Customer ownership
- Server-side validation
- Safe errors
- Sensitive-log exclusion
- Concurrency protection for critical transitions

23.6 Implementation Readiness

The SRS is ready to guide:

- Angular routes, components, guards, and services
- .NET 8 APIs and application use cases
- Domain entities and transition methods
- EF Core models and migrations
- Ocelot routing
- Automated tests
- Demonstration planning

Implementation must not silently resolve the items listed in Section 21. Those decisions require approval or must remain visibly configurable and limited.

23.7 Scope Exclusions

The current MVP does not include:

- Real payment gateway or settlement
- Claim settlement
- Policy renewal
- Fraud detection
- External insurance integrations
- Manual underwriting role or queue
- Multi-level Claim investigation
- Advanced notification infrastructure
- Enterprise document management
- Advanced analytics or BI
- RabbitMQ, Kafka, or Event Bus
- CQRS or Saga orchestration
- Distributed transactions
- Kubernetes, Service Mesh, or production cloud orchestration

23.8 Final Baseline

```
Customer
-> Register and Authenticate
-> Manage Profile
-> Register Vacation Property
-> Select Coverage
-> System Calculates Risk
-> System Calculates Premium
-> Quote Generated
-> Policy Created in Pending
-> Policy Becomes Active under Approved Conditions
-> Customer Submits Claim
-> Claim Becomes Submitted
-> Claims Manager Begins Review
-> Claim Becomes Under Review
-> Claims Manager Approves or Rejects
-> Customer Views Final Outcome
```

This SRS is the consolidated authoritative baseline for implementation and testing of the one-week Vacation Home Insurance Management System capstone.